LAB 7: Fuzzing for Input Validation Bugs

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SECTION : D

1 . Run the buggy application .py



2 .Run the test_processor.py :

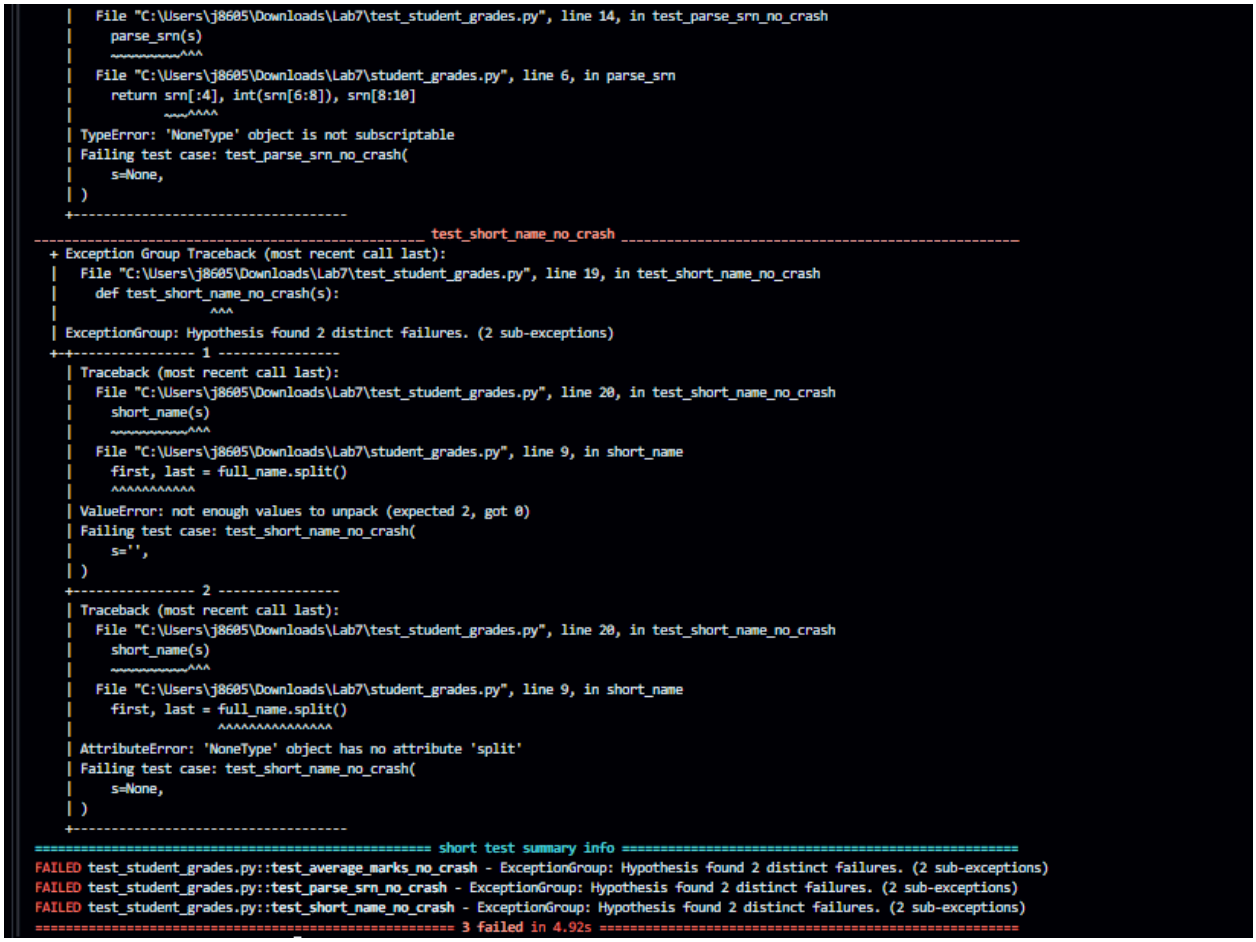
-> screenshot of test cases failing



```
PS C:\Users\j8605\Downloads\Lab7> python -m pytest test_student_grades.py -v
============================================= test session starts =============================================
platform win32 -- Python 3.14.0, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\j8605\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\j8605\Downloads\Lab7
plugins: hypothesis-6.170.15
collected 3 items

test_student_grades.py::test_average_marks_no_crash FAILED                                           [ 33%]
test_student_grades.py::test_parse_srn_no_crash FAILED                                               [ 66%]
test_student_grades.py::test_short_name_no_crash FAILED                                              [100%]

================================================== FAILURES ===================================================
_____________________________________________ test_average_marks_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\j8605\Downloads\Lab7\test_student_grades.py", line 7, in test_average_marks_no_crash
  |     def test_average_marks_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\j8605\Downloads\Lab7\test_student_grades.py", line 8, in test_average_marks_no_crash
    |     average_marks(s)
    |     ~~~~~~~~~~~~~~^^^
    |   File "C:\Users\j8605\Downloads\Lab7\student_grades.py", line 2, in average_marks
    |     marks = [int(m) for m in marks_csv.split(',')]
    |              ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_average_marks_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\j8605\Downloads\Lab7\test_student_grades.py", line 8, in test_average_marks_no_crash
    |     average_marks(s)
    |     ~~~~~~~~~~~~~~^^^
    |   File "C:\Users\j8605\Downloads\Lab7\student_grades.py", line 2, in average_marks
    |     marks = [int(m) for m in marks_csv.split(',')]
    |                            ^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_average_marks_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________________________ test_parse_srn_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\j8605\Downloads\Lab7\test_student_grades.py", line 13, in test_parse_srn_no_crash
  |     def test_parse_srn_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\j8605\Downloads\Lab7\test_student_grades.py", line 14, in test_parse_srn_no_crash
    |     parse_srn(s)
    |     ~~~~~~~~~^^^
    |   File "C:\Users\j8605\Downloads\Lab7\student_grades.py", line 6, in parse_srn
    |     return srn[:4], int(srn[6:8]), srn[8:10]
    |                  ~~~^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_parse_srn_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\j8605\Downloads\Lab7\test_student_grades.py", line 14, in test_parse_srn_no_crash
```



3. Fix the buggy application python script:



4 .Re-Run the test_processor.py wrt to fixed processor and take the ss of the output:

